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(54) **AEROSOL GENERATING ARTICLE**

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**A24D 3/17** (2020.01)

**A24C 5/01** (2020.01)

(52) **U.S. Cl.**

CPC ..... **A24D 1/20** (2020.01); **A24D 3/17** (2020.01); **A24C 5/01** (2020.01)

(58) **Field of Classification Search**

CPC .... **A24B 3/14**; **A24C 5/01**; **A24D 1/20**; **A24F 40/20**; **A24F 40/465**

See application file for complete search history.

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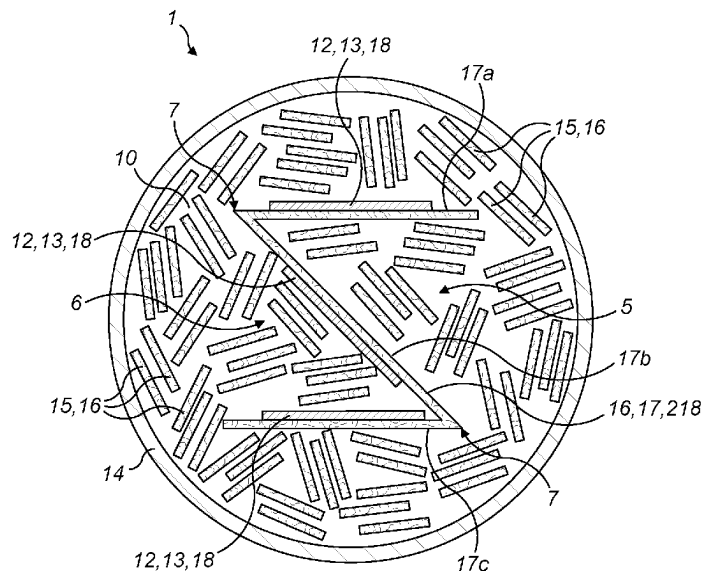
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(57) **ABSTRACT**

An aerosol generating article includes a plurality of elongate first strips having an aerosol generating material; a plurality of elongate second strips having an inductively heatable susceptor material; and at least one elongate carrier strip to which each of the plurality of elongate second strips is attached. The at least one elongate carrier strip is folded along one or more fold lines to form two or more elongate carrier regions, and one or more of the plurality of elongate second strips is attached to each elongate carrier region. The elongate first strips, the elongate second strips and the at least one elongate carrier strip are arranged to form a rod-shaped aerosol generating article.

**19 Claims, 10 Drawing Sheets**



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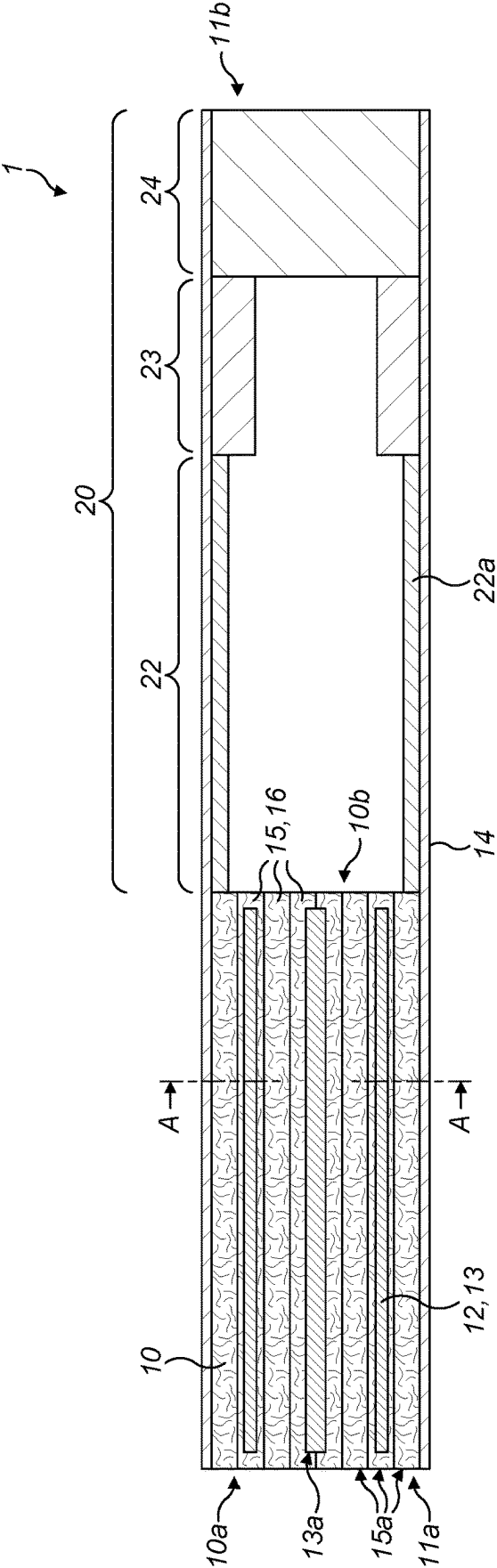


FIG. 1a

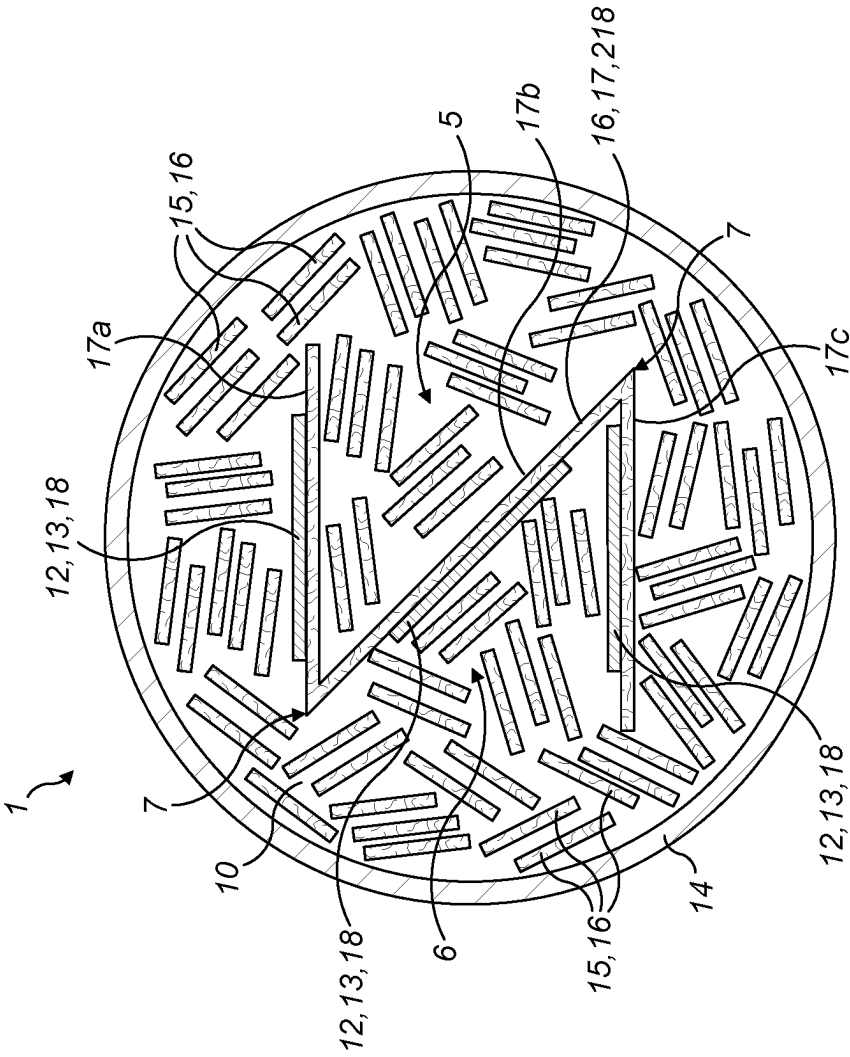


FIG. 1b

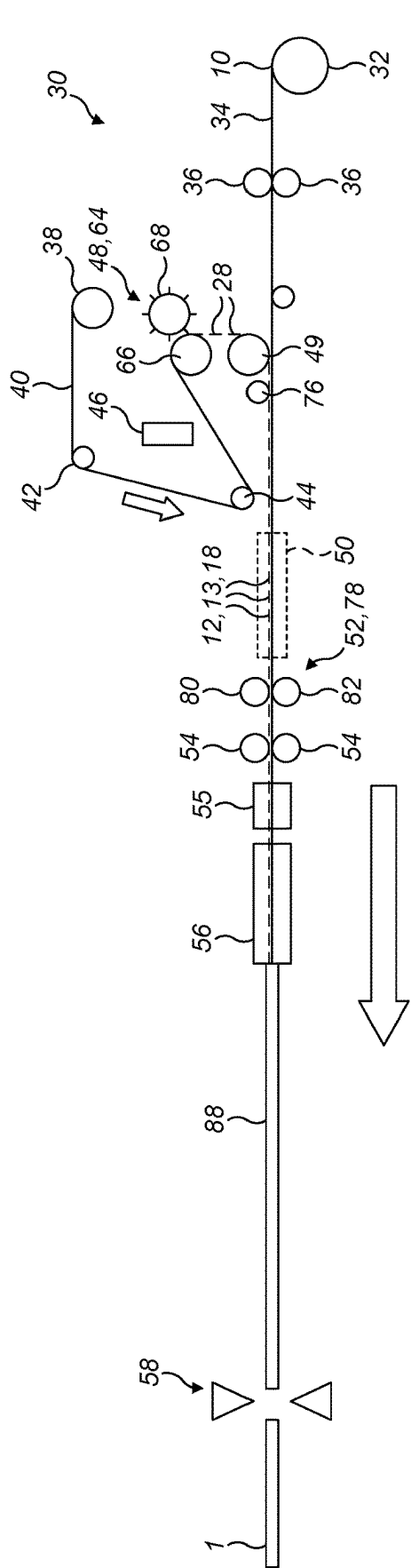


FIG. 2a

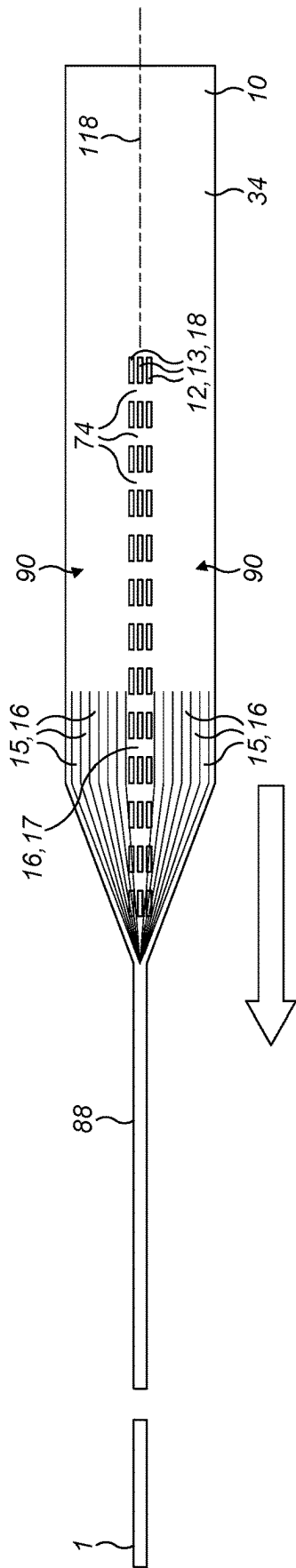
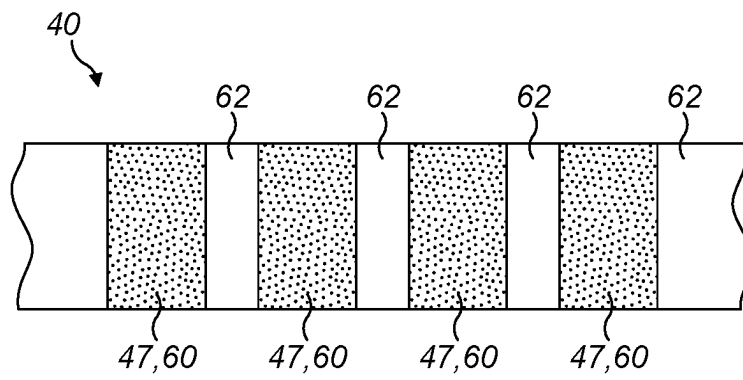


FIG. 2b

**FIG. 3**

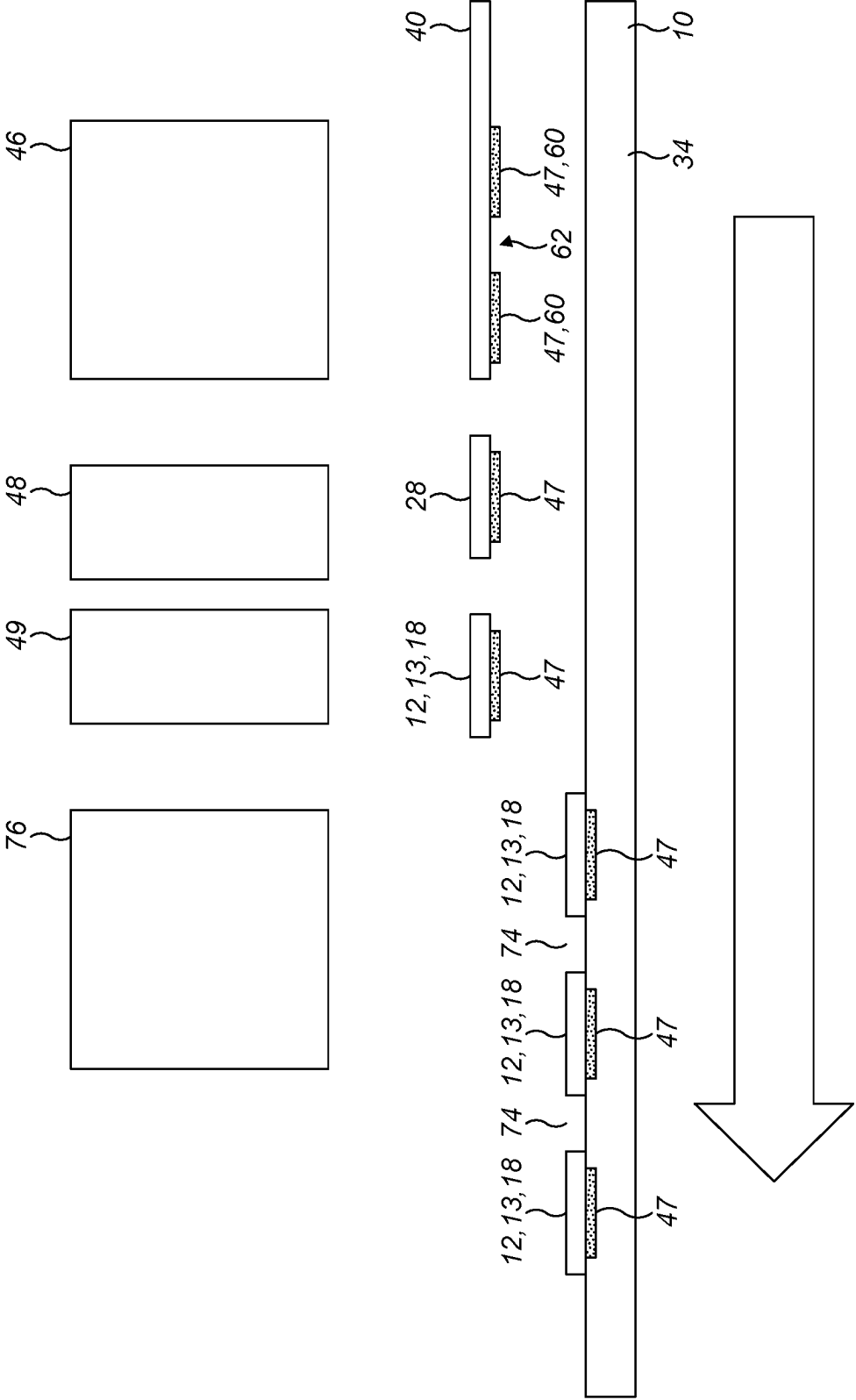


FIG. 4

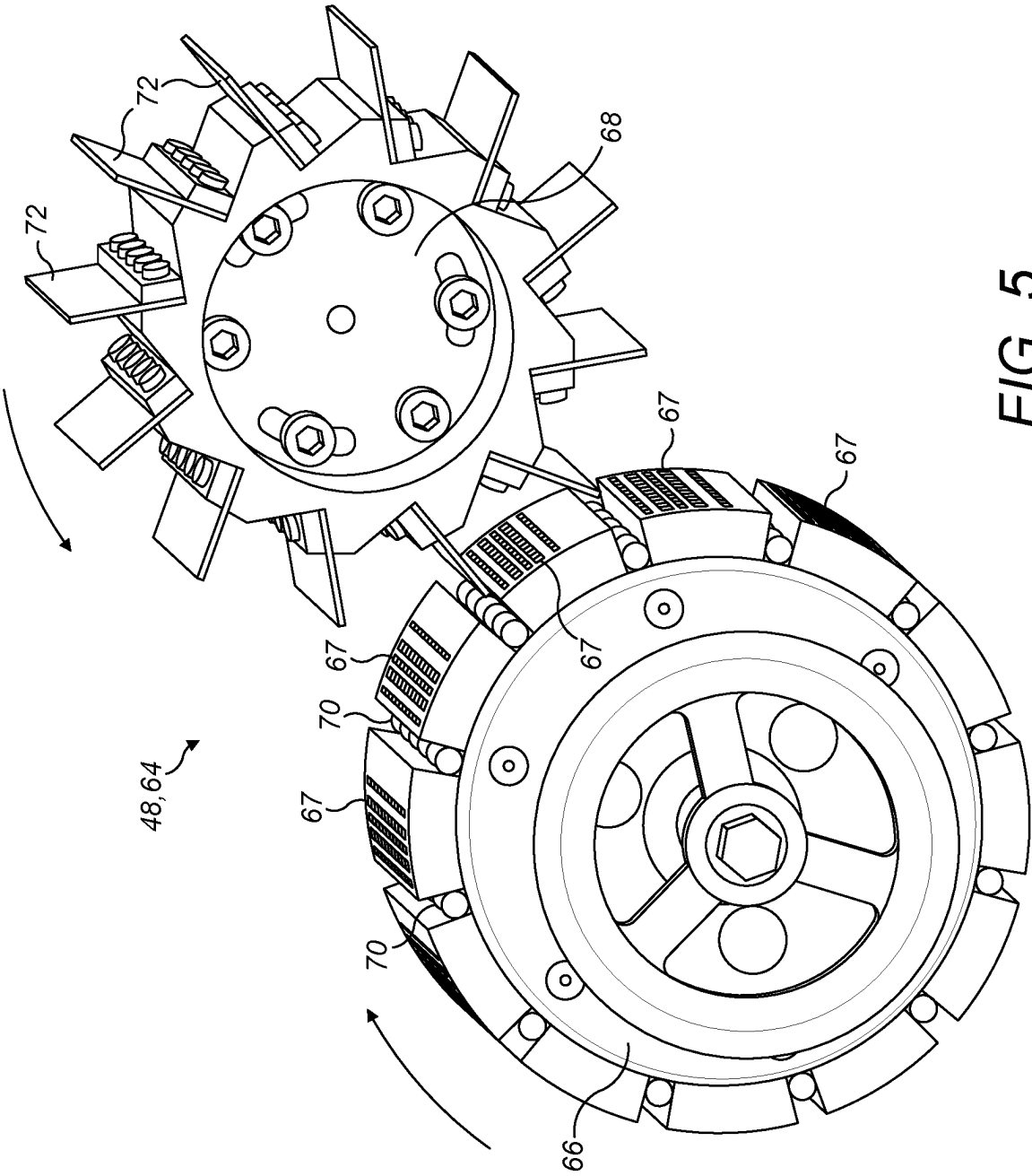


FIG. 5



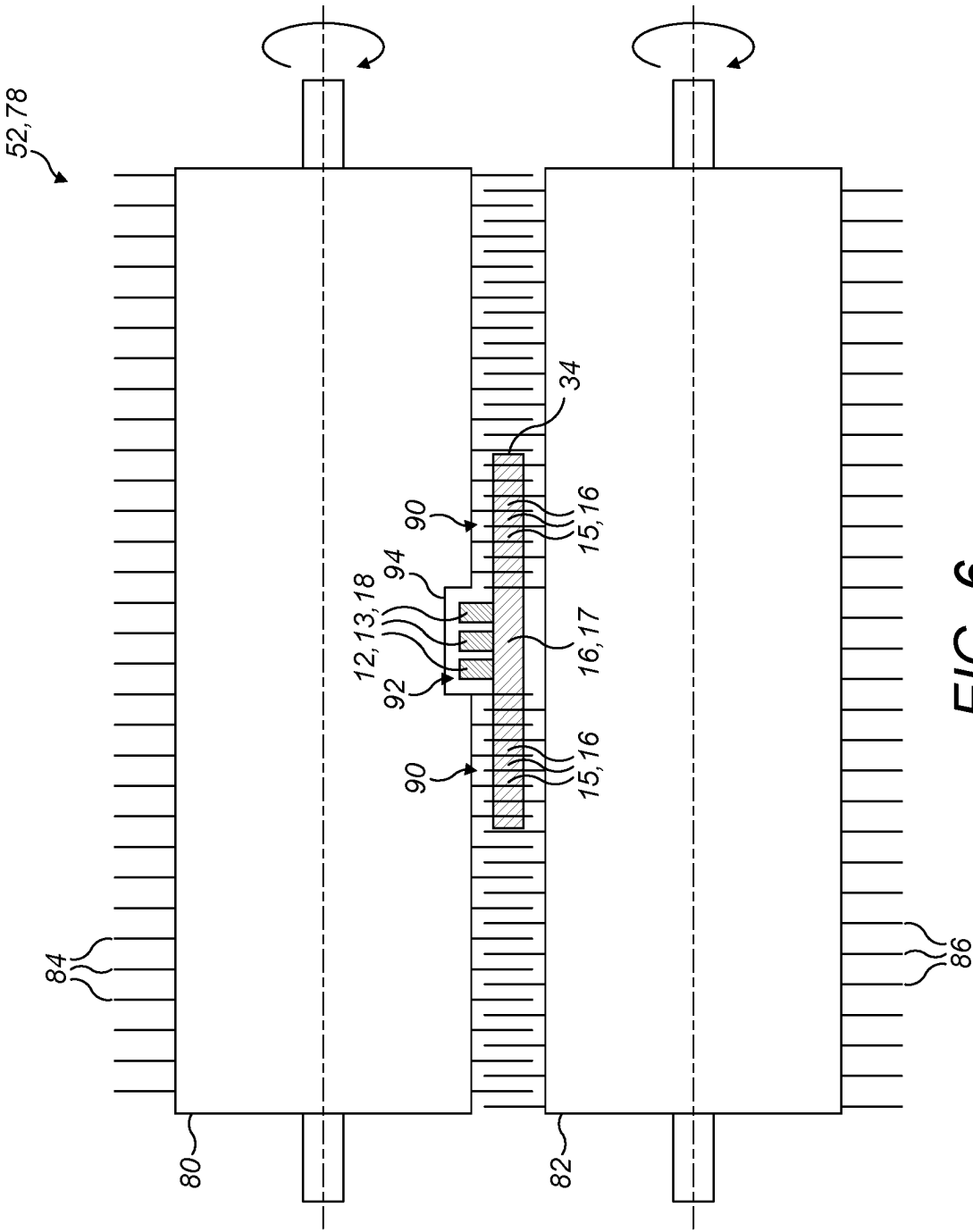


FIG. 6

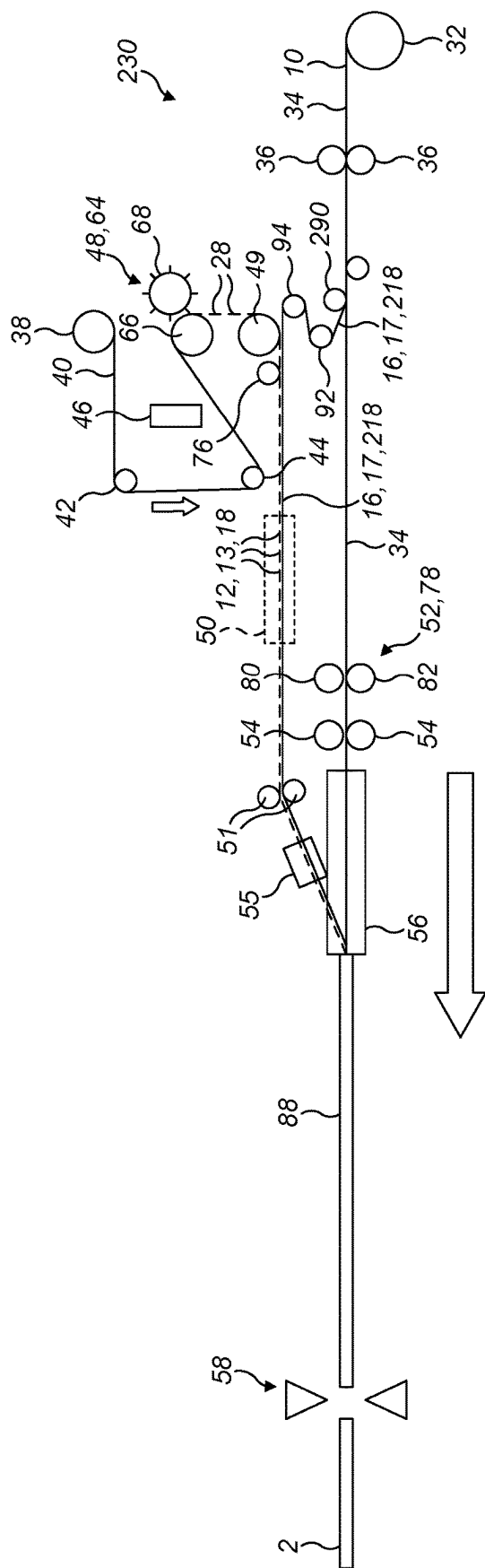


FIG. 7a

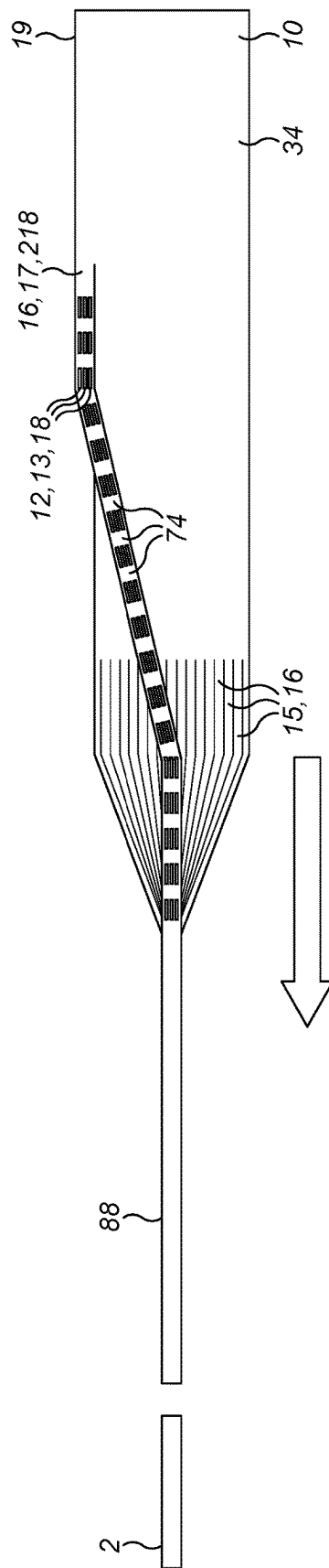
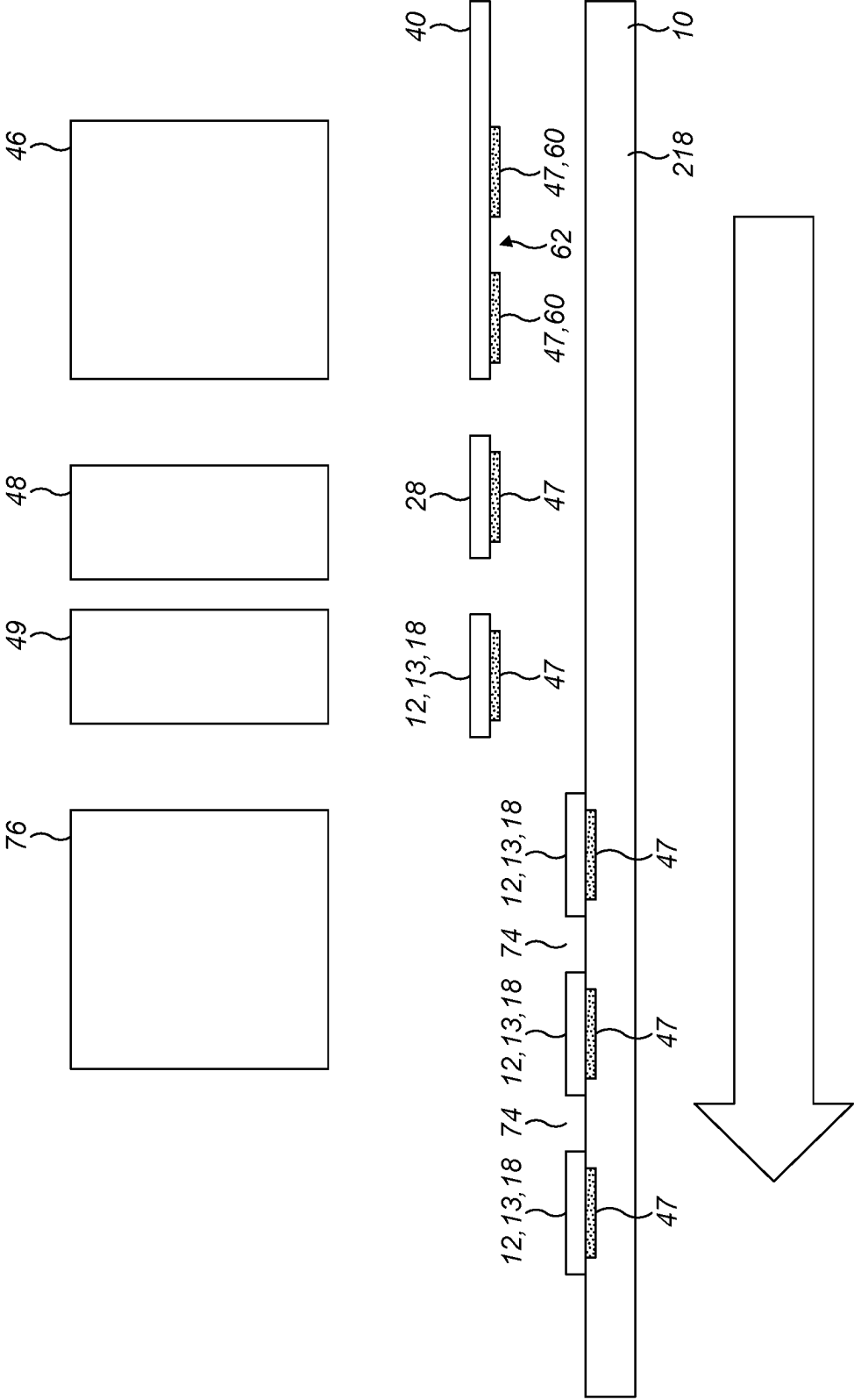


FIG. 7b



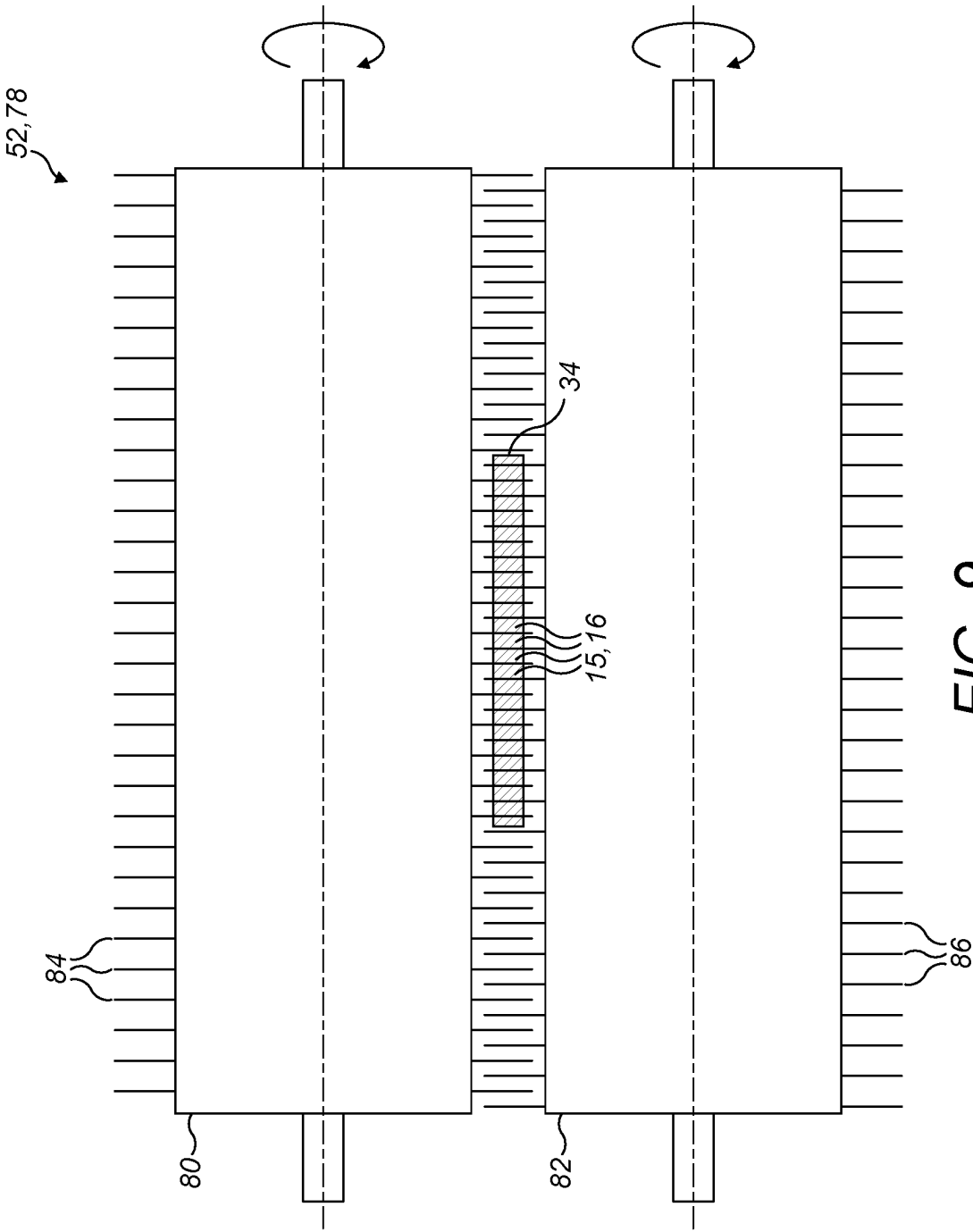


FIG. 9

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**AEROSOL GENERATING ARTICLE****CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/EP2021/075603, filed Sep. 17, 2021, published in English, which claims priority to European Application No. 20197176.9, filed Sep. 21, 2020, the disclosures of which are incorporated herein by reference.

**TECHNICAL FIELD**

The present disclosure relates generally to aerosol generating articles, and more particularly to an aerosol generating article for use with an aerosol generating device for heating the aerosol generating article to generate an aerosol for inhalation by a user. The present disclosure is particularly applicable to aerosol generating articles for use with a portable (hand-held) aerosol generating device.

**TECHNICAL BACKGROUND**

The popularity and use of reduced-risk or modified-risk devices (also known as aerosol generating devices or vapour generating devices) has grown rapidly in recent years as an alternative to the use of traditional tobacco products. Various devices and systems are available that heat or warm aerosol generating substances to generate an aerosol for inhalation by a user.

A commonly available reduced-risk or modified-risk device is the heated substrate aerosol generating device, or so-called heat-not-burn device. Devices of this type generate an aerosol or vapour by heating an aerosol generating substrate to a temperature typically in the range 150° C. to 300° C. Heating the aerosol generating substrate to a temperature within this range, without burning or combusting the aerosol generating substrate, generates a vapour which typically cools and condenses to form an aerosol for inhalation by a user of the device.

Currently available aerosol generating devices can use one of a number of different approaches to provide heat to the aerosol generating substrate. One such approach is to provide an aerosol generating device which employs an induction heating system. In such a device, an induction coil is provided in the device and an inductively heatable susceptor is provided to heat the aerosol generating substrate. Electrical energy is supplied to the induction coil when a user activates the device which in turn generates an alternating electromagnetic field. The susceptor couples with the electromagnetic field and generates heat which is transferred, for example by conduction, to the aerosol generating substrate and an aerosol is generated as the aerosol generating substrate is heated.

The characteristics of the aerosol generated by the aerosol generating device are dependent upon a number of factors, including the construction of the aerosol generating article used with the aerosol generating device. There is, therefore, a desire to provide an aerosol generating article which enables the characteristics of the aerosol generated during use of the article to be optimised. There is also a general desire to provide an aerosol generating article which can be mass-produced easily and consistently.

**SUMMARY OF THE DISCLOSURE**

According to a first aspect of the present disclosure, there is provided an aerosol generating article comprising:

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a plurality of elongate first strips comprising an aerosol generating material;

a plurality of elongate second strips comprising an inductively heatable susceptor material; and

at least one elongate carrier strip to which each of the plurality of elongate second strips is attached;

wherein:

the at least one elongate carrier strip is folded along one or more fold lines to form two or more elongate carrier regions, and one or more of the plurality of elongate second strips is attached to each elongate carrier region, and

the elongate first strips, the elongate second strips and the at least one elongate carrier strip are arranged to form a rod-shaped aerosol generating article.

The aerosol generating article is for use with an aerosol generating device for heating the aerosol generating material, without burning the aerosol generating material, to volatilise at least one component of the aerosol generating material and thereby generate a heated vapour which cools and condenses to form an aerosol for inhalation by a user of the aerosol generating device. The aerosol generating device is a hand-held, portable, device.

In general terms, a vapour is a substance in the gas phase at a temperature lower than its critical temperature, which means that the vapour can be condensed to a liquid by increasing its pressure without reducing the temperature, whereas an aerosol is a suspension of fine solid particles or liquid droplets, in air or another gas. It should, however, be noted that the terms ‘aerosol’ and ‘vapour’ may be used interchangeably in this specification, particularly with regard to the form of the inhalable medium that is generated for inhalation by a user.

The combination of elongate first strips (aerosol generating strips) and a plurality of elongate second strips (susceptor strips) in the aerosol generating article provides effective heat transfer from the elongate second strips to the elongate first strips during use of the aerosol generating article in an aerosol generating device. By providing an elongate carrier strip with one or more carrier regions to which the elongate second strips are attached, the positioning of the elongate second strips relative to the elongate first strips is facilitated, and this further ensures uniform and effective heat transfer from the elongate second strips to the elongate first strips. Effective and uniform heating of the elongate first strips and, thus, reliable vapour generation is thereby achieved. Aerosol generating articles according to the present disclosure can also be manufactured efficiently, and mass produced with relative ease.

The one or more fold lines may extend substantially in the longitudinal direction of the aerosol generating article. Thus, the elongate second strips correspondingly extend in the longitudinal direction of the aerosol generating article. This may facilitate uniform heating of the elongate first strips, which may also extend substantially in the longitudinal direction. It may also facilitate manufacture of the aerosol generating article.

One or more of the elongate second strips may be adhered to each elongate carrier region. A secure connection between each elongate second strip and the corresponding elongate carrier region is thereby achieved. This may be particularly advantageous during manufacture of the aerosol generating article, for example during folding of the elongate carrier strip along the one or more fold lines, and may ensure that the elongate second strips do not become detached from the corresponding elongate carrier regions during folding of the elongate carrier strip.

The at least one elongate carrier strip may be a Z-shaped carrier strip. The Z-shaped carrier strip may define three elongate carrier regions. This shape may facilitate heat transfer from the plurality of elongate second strips to the plurality of elongate first strips, thereby ensuring that the plurality of elongate first strips are uniformly heated and avoiding hot and cold spots in the aerosol generating article.

Each of the plurality of elongate first strips may have a width which is less than a width of each of the plurality of elongate second strips. This may facilitate vapour generation by allowing a large quantity of elongate first strips to be provided in the aerosol generating article. Manufacture of the aerosol generating article may also be facilitated.

Each of the plurality of elongate second strips may have a width which is less than a width of the corresponding elongate carrier region to which the elongate second strip is attached. This may help to ensure that the elongate second strips are not folded or otherwise deformed during folding of the elongate carrier strip along the one or more fold lines.

In some embodiments, the at least one elongate carrier strip may define at least first and second regions within the cross-section of the aerosol generating article. The first and second regions may both include a plurality of the elongate first strips. This may facilitate vapour generation by allowing a large quantity of elongate first strips to be provided in the first and second regions.

Each of the plurality of elongate first strips may have a distal end and each of the plurality of elongate second strips may have a distal end. The distal ends of the elongate first strips may form a distal end of the aerosol generating article. The distal ends of the elongate second strips may be positioned inwardly from the distal ends of the elongate first strips. For example, a length of each of the elongate second strips may be less than a length of each of the elongate first strips. With this arrangement, the distal ends of the elongate second strips (susceptor strips) are not visible at the distal end of the aerosol generating article and this may improve the user acceptance of the aerosol generating article. Furthermore, because the elongate second strips (susceptor strips) are fully embedded in the elongate first strips (aerosol generating strips), this may allow an aerosol or vapour to be generated more effectively because the elongate second strips are fully surrounded by the elongate first strips and, therefore, heat transfer from the elongate second strips to the elongate first strips is maximised.

A length of the at least one elongate carrier strip may be equal to a length of each of the elongate first strips. This may facilitate manufacture of the aerosol generating article.

The at least one elongate carrier strip may comprise an aerosol generating material. This may facilitate manufacture of the aerosol generating article and may also allow a maximum amount of vapour to be generated during use of the aerosol generating article due to heating of both the plurality of elongate first strips and the at least one elongate carrier strip by heat transferred from the elongate second strips.

The elongate first strips may have a plurality of different orientations within the cross-section of the rod-shaped aerosol generating article. This may help to ensure a uniform heat transfer from the elongate second strips to the elongate first strips and, thus, allow a maximum amount of vapour to be generated during use of the aerosol generating article.

Each of the plurality of elongate second strips may have a thickness between 1.0  $\mu\text{m}$  and 500  $\mu\text{m}$ , possibly between 10  $\mu\text{m}$  and 100  $\mu\text{m}$ . Each of the plurality of elongate second strips may have a thickness of 50  $\mu\text{m}$ . Elongate second strips (susceptor strips) having these thickness dimensions may be

particularly suitable for being inductively heated during use of the aerosol generating article and may also facilitate manufacture of the aerosol generating article.

Each of the plurality of elongate first strips may have a length between 5.0 mm and 50 mm, possibly between 10 mm and 30 mm. Each of the plurality of elongate first strips may have a length of 20 mm.

Each of the plurality of elongate first strips may have a thickness between 50  $\mu\text{m}$  and 500  $\mu\text{m}$ , possibly between 150  $\mu\text{m}$  and 300  $\mu\text{m}$ . Each of the plurality of elongate first strips may have a thickness of 220  $\mu\text{m}$ .

Each of the plurality of elongate first strips may have a width of between approximately 0.1 mm and 5.0 mm, possibly between 0.5 mm and 2.0 mm. Each of the plurality of elongate first strips may have a width of 1.0 mm. These width dimensions ensure that the aerosol generating article contains an optimum number of elongate first strips to allow uniform airflow through the aerosol generating article and the generation of an acceptable quantity of vapour or aerosol. If the width of the elongate first strips is too low, the strength of the strips may be reduced and, consequently, mass production of aerosol generating articles may become difficult.

The inductively heatable susceptor material may comprise a metal. The metal is typically selected from the group consisting of stainless steel and carbon steel. The inductively heatable susceptor material could, however, comprise any suitable material including one or more, but not limited, of aluminium, iron, nickel, stainless steel, carbon steel, and alloys thereof, e.g. Nickel Chromium or Nickel Copper. With the application of an electromagnetic field in its vicinity during use of the aerosol generating article in an aerosol generating device, the elongate second strips (susceptor strips) may generate heat due to eddy currents and magnetic hysteresis losses resulting in a conversion of energy from electromagnetic to heat.

The aerosol generating material may be any type of solid or semi-solid material. Example types of aerosol generating solids include powder, granules, pellets, shreds, strands, particles, gel, strips, loose leaves, cut leaves, cut filler, porous material, foam material or sheets. The aerosol generating material may comprise plant derived material and in particular, may comprise a tobacco material. It may advantageously comprise reconstituted tobacco, for example including tobacco and any one or more of cellulose fibres, tobacco stalk fibres and inorganic fillers such as  $\text{CaCO}_3$ .

Consequently, the aerosol generating device with which the aerosol generating articles are intended for use may be referred to as a "heated tobacco device", a "heat-not-burn tobacco device", a "device for vaporising tobacco products", and the like, with this being interpreted as a device suitable for achieving these effects. The features disclosed herein are equally applicable to devices which are designed to vaporise any aerosol generating substrate.

The aerosol generating article may be circumscribed by a paper wrapper.

The aerosol generating article may be formed substantially in the shape of a stick, and may broadly resemble a cigarette, having a tubular region with an aerosol generating substrate arranged in a suitable manner. The aerosol generating article may include a filter segment, for example comprising cellulose acetate fibres, at a proximal end of the aerosol generating article. The filter segment may constitute a mouthpiece filter and may be in coaxial alignment with an aerosol generating substrate constituted by the plurality of elongate first strips. One or more vapour collection regions, cooling regions, and other structures may also be included in

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some designs. For example, the aerosol generating article may include at least one tubular segment upstream of the filter segment. The tubular segment may act as a vapour cooling region. The vapour cooling region may advantageously allow the heated vapour generated by heating the aerosol generating strips (the elongate first strips and preferably the at least one elongate carrier strip) to cool and condense to form an aerosol with suitable characteristics for inhalation by a user, for example through the filter segment.

The aerosol generating material may comprise an aerosol-former. Examples of aerosol-formers include polyhydric alcohols and mixtures thereof such as glycerine or propylene glycol. Typically, the aerosol generating material may comprise an aerosol-former content of between approximately 5% and approximately 50% on a dry weight basis. In some embodiments, the aerosol generating material may comprise an aerosol-former content of between approximately 10% and approximately 20% on a dry weight basis, and possibly approximately 15% on a dry weight basis.

Upon heating, the aerosol generating material may release volatile compounds. The volatile compounds may include nicotine or flavour compounds such as tobacco flavouring.

#### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1a is a diagrammatic cross-sectional side view of an example of an aerosol generating article;

FIG. 1b is an enlarged diagrammatic cross-sectional view along the line A-A in FIG. 1a;

FIG. 2a is a diagrammatic illustration of a first embodiment of an apparatus and method for manufacturing the aerosol generating article illustrated in FIGS. 1a and 1b;

FIG. 2b is a plan view of an aerosol generating substrate and susceptor strips as the aerosol generating substrate and susceptor strips move in the direction shown by the arrow through the apparatus illustrated in FIG. 2a;

FIG. 3 is a plan view of a section of a continuous web of susceptor material showing adhesive areas and non-adhesive areas;

FIG. 4 is a functional illustration of part of the apparatus and method of FIG. 2a schematically illustrating the formation of susceptor patches and susceptor strips from a continuous web of susceptor material and the application of the susceptor strips to a surface of a continuous web of aerosol generating substrate;

FIG. 5 is a diagrammatic perspective view of a susceptor cutting unit;

FIG. 6 is a diagrammatic illustration of a strip cutting unit of the apparatus of FIG. 2a;

FIG. 7a is a diagrammatic illustration of a second embodiment of an apparatus and method for manufacturing the aerosol generating article illustrated in FIGS. 1a and 1b;

FIG. 7b is a plan view of an aerosol generating substrate and susceptor strips as the aerosol generating substrate and susceptor strips move in the direction shown by the arrow through the apparatus illustrated in FIG. 7a;

FIG. 8 is a functional illustration of part of the apparatus and method of FIG. 7a schematically illustrating the formation of susceptor patches and susceptor strips from a continuous web of susceptor material and the application of the susceptor strips to a surface of a continuous strip of aerosol generating substrate; and

FIG. 9 is a diagrammatic illustration of a strip cutting unit of the apparatus of FIG. 7a.

#### DETAILED DESCRIPTION OF EMBODIMENTS

Embodiments of the present disclosure will now be described by way of example only and with reference to the accompanying drawings.

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Referring to FIGS. 1a and 1b, there is shown an example of an aerosol generating article 1 for use with an aerosol generating device that comprises an induction heating system to inductively heat the aerosol generating article and thereby generate an aerosol for inhalation by a user of the device. Such devices are known in the art and will not be described in further detail in this specification. The aerosol generating article 1 is elongate, having a distal end 11a and a proximal end (or mouth end) 11b, and is substantially cylindrical. The circular cross-section facilitates handling of the article 1 by a user and insertion of the article 1 into a cavity or heating compartment of an aerosol generating device.

The aerosol generating article 1 comprises an aerosol generating substrate 10 having first and second ends 10a, 10b and an inductively heatable susceptor 12. The aerosol generating substrate 10 and the inductively heatable susceptor 12 are positioned in, and enclosed by, a wrapper 14. The wrapper 14 comprises a material which is substantially non-electrically conductive and non-magnetically permeable. In the illustrated example, the wrapper 14 is a paper wrapper and may comprise cigarette paper.

The aerosol generating article 1 may have a total length, measured between the distal end 11a and the proximal (mouth) end 11b, between 30 mm and 100 mm, possibly between 50 mm and 70 mm. The aerosol generating article 1 may have a total length of approximately 55 mm. The aerosol generating substrate 10 may have a total length, measured between the first and second ends 10a, 10b, between 5.0 mm and 50 mm, possibly between 10 mm and 30 mm. The aerosol generating substrate 10 may have a total length of approximately 20 mm. The aerosol generating article 1 may have a diameter between 5.0 mm and 10 mm, possibly between 6.0 mm and 8.0 mm. The aerosol generating article 1 may have a diameter of approximately 7.0 mm.

The aerosol generating substrate 10 comprises a plurality of elongate first strips 15 comprising an aerosol generating material. The plurality of elongate first strips 15 constitute aerosol generating strips 16 and are substantially oriented in a longitudinal direction of the aerosol generating article 1. The elongate first strips 15 are typically foldless in the longitudinal direction to ensure that the air flow route is not interrupted and that a uniform air flow through the article 1 can be achieved.

The inductively heatable susceptor 12 comprises a plurality of elongate second strips 13 comprising an inductively heatable susceptor material. The plurality of elongate second strips 13 constitute susceptor strips 18 and are also substantially oriented in the longitudinal direction of the aerosol generating article 1. The elongate second strips 13 are foldless in the longitudinal direction to prevent hot spots in the aerosol generating substrate 10. As can be clearly seen in FIG. 1b, each of the elongate first strips 15 has a width which is less than a width of each elongate second strip 13.

The aerosol generating article 1 comprises an elongate carrier strip 17 oriented substantially in the longitudinal direction of the aerosol generating article 1. The elongate carrier strip 17 has a plurality of fold lines 7 which extend substantially in the longitudinal direction of the aerosol generating article 1 and the elongate carrier strip 17 is folded along the fold lines 7 to define elongate carrier regions 17a, 17b, 17c. In the illustrated example, the elongate carrier strip 17 is configured as a substantially Z-shaped carrier strip 17 with two fold lines 7, and defines three separate elongate carrier regions 17a, 17b, 17c. It will, however, be understood that other configurations of the elongate carrier strip 17 are

entirely within the scope of the present disclosure, provided that the elongate carrier strip 17 has at least two elongate carrier regions 17a, 17b. By way of example, an elongate carrier strip 17 having two elongate carrier regions 17a, 17b could be configured as a V-shaped carrier strip 17 or as an L-shaped carrier strip 17.

The elongate carrier strip 17 comprises an aerosol generating material and, thus, also constitutes an aerosol generating strip 16. The elongate carrier strip 17 has the same length as the elongate first strips 15, and thus the aerosol generating strips 16 within the aerosol generating article 1 all have the same length.

The elongate second strips 13 are adhered to the elongate carrier strip 17. More particularly, and as can be clearly seen in FIG. 1b, one elongate second strip 13 is adhered to each elongate carrier region 17a, 17b, 17c. In the illustrated example, each of the plurality of elongate second strips 13 has a width which is less than a width of the corresponding elongate carrier region 17a, 17b, 17c to which the elongate second strip 13 is attached.

The elongate first strips 15, the elongate second strips 13 and the elongate carrier strip 17 are arranged to form a substantially rod-shaped aerosol generating article 1 and the elongate first strips 15 can be randomly distributed throughout the cross-section of the rod-shaped aerosol generating article 1 such that they have a plurality of different orientations within the cross-section of the aerosol generating article 1. Although not apparent from FIG. 1b, a sufficient number of elongate first strips 15 are provided to substantially fill the cross-section of the aerosol generating substrate 10, and it will be understood that a smaller number of elongate first strips 15 are shown merely for illustration purposes. The elongate second strips 13 and the elongate carrier strip 17 are positioned roughly centrally within the cross-section of the aerosol generating substrate 10, and hence the aerosol generating article 1. Such an arrangement may, for example, help to ensure that there is uniform heat transfer from the elongate second strips 13 to the elongate first strips 15.

As best seen in FIG. 1b, the centrally positioned elongate carrier strip 17 and the elongate second strips 13 adhered to the elongate carrier regions 17a, 17b, 17c can be considered to define at least first and second regions 5, 6 within the cross-section of the aerosol generating substrate 10 and, hence, within the cross-section of the aerosol generating article 1. The first and second regions 5, 6 both include a plurality of elongate first strips 15.

As best seen in FIG. 1a, each of the plurality of elongate first strips 15 has a distal end 15a and each of the plurality of elongate second strips 13 has a distal end 13a. The distal ends 15a of the elongate first strips 15 form the first end 10a of the aerosol generating substrate 10 and, correspondingly, the distal end 11a of the aerosol generating article 1. The elongate second strips 13 are shorter than the elongate first strips 15 and the elongate carrier strip 17. The distal ends 13a of the elongate second strips 13 are positioned inwardly from the distal ends 15a of the elongate first strips 15. The distal ends 13a of the elongate second strips 13 (i.e., the susceptor 12) are, therefore, not visible at the distal end 11a of the aerosol generating article 1.

The aerosol generating article 1 comprises a mouthpiece segment 20 positioned downstream of the aerosol generating substrate 10. The aerosol generating substrate 10 and the mouthpiece segment 20 are arranged in coaxial alignment inside the wrapper 14 to hold the components in position to form the rod-shaped aerosol generating article 1.

In the illustrated embodiment, the mouthpiece segment 20 comprises the following components arranged sequentially and in co-axial alignment in a downstream direction, in other words from the distal end 11a to the proximal (mouth) end 11b of the aerosol generating article 1: a cooling segment 22, a center hole segment 23 and a filter segment 24. The cooling segment 22 comprises a hollow paper tube 22a having a thickness which is greater than the thickness of the paper wrapper 14. The center hole segment 23 may comprise a cured mixture containing cellulose acetate fibres and a plasticizer, and functions to increase the strength of the mouthpiece segment 20. The filter segment 24 typically comprises cellulose acetate fibres and acts as a mouthpiece filter. As heated vapour flows from the aerosol generating substrate 10 towards the proximal (mouth) end 11b of the aerosol generating article 1, the vapour cools and condenses as it passes through the cooling segment 22 and the center hole segment 23 to form an aerosol with suitable characteristics for inhalation by a user through the filter segment 24.

The elongate first strips 15 and the elongate carrier strip 17 typically comprise plant derived material, such as tobacco. The elongate first strips 15 and the elongate carrier strip 17 can advantageously comprise reconstituted tobacco including tobacco and any one or more of cellulose fibres, tobacco stalk fibres and inorganic fillers such as CaCO<sub>3</sub>.

The elongate first strips 15 and the elongate carrier strip 17 typically comprise an aerosol-former such as glycerine or propylene glycol. Typically, the elongate first strips 15 and the elongate carrier strip 17 comprise an aerosol-former content of between approximately 5% and approximately 50% on a dry weight basis. Upon heating, the elongate first strips 15 and the elongate carrier strip 17 release volatile compounds possibly including nicotine or flavour compounds such as tobacco flavouring.

When a time varying electromagnetic field is applied in the vicinity of the elongate second strips 13 during use of the article 1 in an aerosol generating device, heat is generated in the elongate second strips 13 due to eddy currents and magnetic hysteresis losses. The heat is transferred from the elongate second strips 13 to the elongate first strips 15 and the elongate carrier strip 17 to heat the elongate first strips 15 and the elongate carrier strip 17 without burning them to release one or more volatile compounds and thereby generate a vapour. As a user inhales through the filter segment 24, the heated vapour is drawn in a downstream direction through the article 1 from the first end 10a of the aerosol generating substrate 10 towards the second end 10b of the aerosol generating substrate 10, and towards the filter segment 24. As noted above, as the heated vapour flows through the cooling segment 22 and the center hole segment 23 towards the filter segment 24, the heated vapour cools and condenses to form an aerosol with suitable characteristics for inhalation by a user through the filter segment 24.

Apparatus 30, 230 and methods suitable for manufacturing aerosol generating articles according to the present disclosure, such as the aerosol generating article 1 described above with reference to FIGS. 1a and 1b, will now be described.

Manufacture of Aerosol Generating Articles: Embodiment 1

Referring to FIG. 2a, there is shown a diagrammatic illustration of an apparatus 30 and method for manufacturing the aerosol generating article 1 described above with reference to FIGS. 1a and 1b. FIG. 2b is a plan view of an aerosol generating substrate 10 and susceptor strips 18 as they move through the apparatus 30, in the direction of the arrow in FIG. 2b.



The apparatus 30 comprises a substrate supply reel 32 (e.g. a first bobbin) which carries a continuous web 34 of an aerosol generating substrate 10 having a substantially flat surface with a centre line 118 and first feed rollers 36 for controlling the feed of the continuous web 34 of aerosol generating substrate 10. The apparatus 30 may also include a web tension regulator and a web edge control system as will be understood by one of ordinary skill in the art, but these additional components are not essential in the context of the present disclosure and have, therefore, been omitted for the sake of simplicity.

The apparatus 30 comprises a susceptor supply reel 38 (e.g. a second bobbin) which carries a continuous web 40 of susceptor material, feed rollers 42, 44 for controlling the feed of the continuous web 40 of susceptor material, an adhesive applicator unit 46, a first susceptor cutting unit 48 and a second susceptor cutting unit 49.

The apparatus 30 further comprises an optional heater 50, a strip cutting unit 52, feed rollers 54, a strip folding unit 55, a rod forming unit 56, and a rod cutting unit 58.

#### Susceptor Strip Preparation

In operation, a continuous web 34 of aerosol generating substrate 10 is continuously supplied from the substrate supply reel 32. At the same time, a continuous web 40 of susceptor material is continuously supplied from the susceptor supply reel 38, via the feed rollers 42, 44, to the adhesive applicator unit 46. The adhesive applicator unit 46 applies an adhesive 47 to a surface of the continuous web 40 of susceptor material. In the illustrated example, the adhesive applicator unit 46 applies the adhesive 47 to the surface of the continuous web 40 of susceptor material intermittently, and across the full width of the web 40. In this way, discrete adhesive areas 60 (see FIGS. 3 and 4) are formed on the surface of the continuous web 40 of susceptor material, with adhesive-free areas 62 being formed between adjacent adhesive areas 60 in the direction of travel of the continuous web 40 of susceptor material.

The continuous web 40 of susceptor material is supplied from the adhesive applicator unit 46 to the first susceptor cutting unit 48 which continuously cuts the continuous web of susceptor material to form a plurality of susceptor patches 28 (see FIG. 4). The continuous web 40 of susceptor material, and hence the susceptor patches 28, have a width which is substantially less than a width of the continuous web 34 of aerosol generating substrate 10. For example, the continuous web 34 of aerosol generating substrate 10 can have a width of approximately 140 mm whereas the continuous web of susceptor material, and hence the susceptor patches 28, can have a width of between approximately 0.1 mm and 5 mm. In some embodiments, the susceptor patches 28 can have a length of between approximately 5 mm and 50 mm in the direction of travel of the continuous web 40 of susceptor material and can have a thickness of between approximately 1 µm and 500 µm.

In order to minimise soiling of the first susceptor cutting unit 48 by the adhesive 47 applied to the continuous web 40 of susceptor material by the adhesive applicator unit 46, the first susceptor cutting unit 48 cuts the continuous web 40 of susceptor material in the adhesive-free areas 62, that is at positions between the adhesive areas 60 on the surface of the continuous web 40 of susceptor material (see FIG. 4). This can be achieved by synchronising the operation of the first susceptor cutting unit 48 with the movement of the continuous web 40 of susceptor material.

Referring to FIG. 5, the first susceptor cutting unit 48 comprises a rotary cutting unit 64 comprising a support drum 66 and a cutting drum 68. The support drum 66

supports the continuous web 40 of susceptor material around its periphery and includes a plurality of circumferentially spaced recesses 70 around its periphery. The support drum 66 is typically a suction drum and the continuous web 40 of susceptor material and susceptor patches 28 are supported around the periphery of the suction drum by a suction force applied through suction ports 67. The cutting drum 68 includes a plurality of circumferentially spaced cutting elements 72, for example projecting cutting blades, around its periphery and the cutting elements 72 cooperate with (e.g., extend into) the circumferentially spaced recesses 70 during synchronised rotation of both the support drum 66 and the cutting drum 68 in opposite directions as shown by the arrows in FIG. 5. This results in continuous shear cutting of the continuous web 40 of susceptor material to form a plurality of susceptor patches 28.

The second susceptor cutting unit 49 is positioned downstream of the first susceptor cutting unit 48 and cuts each susceptor patch 28 into a plurality of susceptor strips 18, in the illustrated embodiment three susceptor strips 18 as best seen in FIG. 2b. As will become apparent from the description below, each susceptor strip 18 corresponds to an elongate second strip 13 (i.e., the elongate susceptor 12) in the finished aerosol generating article 1 described above with reference to FIGS. 1a and 1b.

#### Susceptor Strip Application

The susceptor strips 18 provided by the second susceptor cutting unit 49 can be applied to the surface of the continuous web 34 of aerosol generating substrate 10 so that there is a constant and predetermined spacing 74 between the edges of each successive set of susceptor strips 18 in the longitudinal direction (i.e., the direction of travel) of the continuous web 34, for example as shown in FIGS. 2b and 4, and so that there is a small lateral spacing between the susceptor strips 18 in the transverse (width) direction of the continuous web 34 as shown in FIG. 2b. The constant and predetermined spacing 74 between the edges of the susceptor strips 18 may be between 1.0 mm and mm. The constant and predetermined spacing 74 may be approximately 5.0 mm. In order to generate the constant and predetermined spacing 74 between the edges of the adjacent susceptor strips 18 formed by cutting consecutive susceptor patches 28, the first susceptor cutting unit 48 permits relative movement between the continuous web of susceptor material and the support drum 66 for a predetermined period of time immediately after the continuous web 40 of susceptor material carried by the support drum 66 has been cut by the cutting drum 68 to form a susceptor patch 28. This relative movement allows the continuous web 40 of susceptor material to remain stationary or to travel at a reduced speed for a short period of time after a susceptor patch 28 has been cut from the continuous web 40 of susceptor material. The relative movement between the continuous web 40 of susceptor material and the support drum 66 can be achieved by, for example, reducing the suction force applied to the continuous web 40 of susceptor material by the support drum 66, whilst at the same time maintaining an adequate suction force between the already cut susceptor patches 28 and the support drum 66 to ensure that there is no relative movement between the susceptor patches 28 and the support drum 66. In this way, a susceptor patch 28 that has been cut from the continuous web 40 of susceptor material by the first susceptor cutting unit 48 is conveyed for a short period of time at a greater speed than the continuous web 40 of susceptor material from which the susceptor patch 28 has been cut, thereby generating the desired constant and predetermined spacing 74 between the edges of adjacent sus-

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ceptor patches 28 and, hence, between the susceptor strips 18 formed by the second susceptor cutting unit 49 by virtue of cutting the spaced susceptor patches 28 supplied by the first susceptor cutting unit 48.

The susceptor strips 18 with the adhesive 47 applied thereto are continuously and consecutively adhered to the flat surface of the continuous web 34 of aerosol generating substrate 10 substantially along the centre line 118. Exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10 are thereby formed on both sides of the susceptor strips 18 (see FIG. 2b) because, as noted above, the continuous web 34 of aerosol generating substrate 10 is substantially wider than the susceptor patches 28 and, hence, the each set of adhered susceptor strips 18. Adjacent sets of susceptor strips 18 are also spaced apart in the direction of travel of the continuous web 34 of aerosol generating substrate 10 by the constant and predetermined spacing 74 between the edges of the susceptor strips 28 that is generated when the susceptor patches 28 are formed in the first susceptor cutting unit 48.

In order to ensure that there is adequate adhesion between the susceptor strips 18 and the substantially flat surface of the continuous web 34 of aerosol generating substrate 10, the susceptor strips 18 can be pressed onto the substantially flat surface by a cam roller 76, shown diagrammatically in FIG. 2a. The rotation of the cam roller 76 is synchronized with the movement of the continuous web 34 of aerosol generating substrate 10 so that a pressing force is applied to consecutive susceptor strips 18, but not to the spaced regions between consecutive susceptor strips 18.

Depending on the properties of the adhesive 47 applied to the continuous web 40 of susceptor material (and hence to the susceptor strips 18) by the adhesive applicator unit 46, the continuous web 34 of aerosol generating substrate 10 and the susceptor strips 18 adhered to the surface thereof can be heated by the optional heater 50. This may help to cure or set the adhesive 47, and thereby ensure a good bond between the susceptor strips 18 and the flat surface of the continuous web 34 of aerosol generating substrate 10. The heating temperature must be carefully selected based on the characteristics of both the aerosol generating substrate 10 and the adhesive 47, to ensure that sufficient heating is achieved to cure or set the adhesive 47, whilst at the same time avoiding or at least minimising the release of volatile components from the aerosol generating substrate 10.

#### Strip Cutting

The continuous web 34 of aerosol generating substrate 10 with the longitudinally spaced sets of susceptor strips 18 adhered to its flat surface is fed to the strip cutting unit 52. The strip cutting unit 52 cuts only the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10, without cutting the susceptor strips 18, to form a plurality of continuous aerosol generating strips 16 alongside the susceptor strips 18. In an embodiment, the strip cutting unit 52 cuts the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10 to form aerosol generating strips 16 having a strip width of approximately 1 mm.

As shown in FIGS. 2a and 6, the strip cutting unit 52 is a rotary cutter unit 78 and comprises first and second cutting drums 80, 82. The first cutting drum 80 includes circumferentially extending first cutting formations 84 and the second cutting drum 82 includes circumferentially extending second cutting formations 86. The first and second cutting formations 84, 86 cooperate (e.g., intermesh) to shear cut the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10 in the direction of travel of the

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continuous web 34 to form the continuous aerosol generating strips 16, and specifically to form the elongate first strips 15 illustrated in FIGS. 1a and 1b.

In order to provide for cutting of only the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10 to form the elongate first strips 15, the first and second cutting drums 80, 82 define therebetween a non-cutting region 92 which accommodates the susceptor strips 18 (i.e., the elongate second strips 13) and the part of the continuous web 34 of aerosol generating substrate 10 to which the susceptor strips 18 are adhered. In the illustrated embodiment, the first cutting drum 80 is formed without the first cutting formations 84 in the non-cutting region 92. Similarly, the second cutting drum 82 is also formed without the second cutting formations 86 in the non-cutting region 92. Furthermore, the first cutting drum 80 includes a circumferentially extending recess 94 in its surface in the non-cutting region 92, so that at least part of the susceptor strips 18 can be accommodated in the circumferentially extending recess 94 during cutting of the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10. It will, thus, be understood that when the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10 are cut to form the elongate first strips 15 by virtue of the cooperation between the first and second cutting formations 84, 86 on the first and second cutting drums 80, 82 respectively, the central portion of the continuous web 34 of aerosol generating substrate 10 that is accommodated in the non-cutting region 92 and that is not cut into strips constitutes the elongate carrier strip 17 described above with reference to FIG. 1b.

#### Strip Folding and Rod Formation

The elongate first strips 15 (i.e., the aerosol generating strips 16) formed by cutting the exposed side regions 90 of the continuous web 34 of aerosol generating substrate 10, the elongate carrier strip 17 and the adhered elongate second strips 13 (i.e., the susceptor strips 18) are conveyed to the strip folding unit 55 which folds the elongate carrier strip 17 along fold lines 7 to form the Z-shaped carrier strip 17 described above having elongate carrier regions 17a, 17b, 17c to which the elongate second strips 13 are adhered. After that, the elongate first strips 15 and the Z-shaped carrier strip 17 with the adhered elongate second strips 13 are conveyed to the rod forming unit 56 where they are formed into a continuous rod 88.

If desired, a continuous sheet of wrapping paper (not shown) can be supplied to the rod forming unit 56 from a supply reel (not shown) or can be supplied to a separate wrapping unit (again from a supply reel) which can be positioned downstream of the rod forming unit 56. As the sheet of wrapping paper is transported and guided through the rod forming unit 56 or the separate wrapping unit, it can be wrapped around the elongate first strips 15, the Z-shaped carrier strip 17 and the adhered elongate second strips 13 so that the continuous rod 88 is circumscribed by a wrapper 14. Rod Cutting

The continuous rod 88 (optionally circumscribed by a wrapper 14) is then transported to the rod cutting unit 58 where it is cut at appropriate positions into predetermined lengths to form multiple aerosol generating articles 1. The aerosol generating articles 1 formed by the rod cutting unit 58 may have a length between 5.0 mm and 50 mm, possibly between 10 mm and 30 mm. The aerosol generating articles 1 formed by the rod cutting unit 58 may have a length of 20 mm. It will be understood that this length corresponds to the length of the aerosol generating substrate 10 described above with reference to FIGS. 1a and 1b. The continuous

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rod **88** is preferably cut repeatedly by the rod cutting unit **58** substantially at a midpoint between the ends of the longitudinally spaced elongate second strips **13** (i.e., the susceptor strips **18**). In this way, the elongate second strips **13** are not cut by the rod cutting unit **58**, thereby reducing wear on the cutting elements. Further, because the elongate second strips **13** are shorter than the aerosol generating strips **16**, the ends of the elongate second strips **13** (i.e., the susceptor strips **18**) are not visible at either end of the aerosol generating articles **1** formed by the rod cutting unit **58**. It will be understood that this type of method is particularly suitable for the mass production of aerosol generating articles **1**. Final Assembly

Further units (not shown) may be arranged downstream of the rod cutting unit **58** and may be configured to provide one or more additional components such as the mouthpiece segment **20** described above and to assemble these with the individual aerosol generating articles **1** formed by the rod cutting unit **56** to form finished aerosol generating articles **1**, for example of the type illustrated in FIG. **1**. In this case, a separate wrapping unit may be provided downstream of the rod cutting unit **58** so that the assembled components can be simultaneously wrapped to form the finished aerosol generating articles **1**. The further units may form part of the apparatus **30** or may be separate, stand-alone, units forming part of a final assembly line.

Manufacture of Aerosol Generating Articles: Embodiment 2

Referring to FIG. **7a**, there is shown a diagrammatic illustration of a second embodiment of an apparatus **230** and method for manufacturing the aerosol generating article **1** described above with reference to FIGS. **1a** and **1b**. FIG. **7b** is a plan view of an aerosol generating substrate **10** and susceptor strips **18** as they move through the apparatus **230**, in the direction of the arrow in FIG. **7b**. The apparatus **230** and method are similar to the apparatus **30** and method described above with reference to FIGS. **2** to **6** and corresponding components will be identified using the same reference numerals.

The apparatus **230** comprises a substrate supply reel **32** (e.g. a first bobbin) which carries a continuous web **34** of an aerosol generating substrate **10** having a substantially flat surface and first feed rollers **36** for controlling the feed of the continuous web **34** of aerosol generating substrate **10**. The apparatus **230** may also include a web tension regulator and a web edge control system as will be understood by one of ordinary skill in the art, but these additional components are not essential in the context of the present disclosure and have, therefore, been omitted for the sake of simplicity.

The apparatus **230** further comprises a rotary cutter unit **290**, for example including a circular cutting knife, which cuts the continuous web **34** of aerosol generating substrate along one edge **19** to separate a continuous strip **218** of aerosol generating substrate from the continuous web **34**. As discussed below, the continuous strip **218** of aerosol generating substrate **10** is subsequently folded along fold lines **7** to form the elongate carrier strip **17** in the finished aerosol generating article **1** described above with reference to FIGS. **1a** and **1b**. The continuous strip **218** of aerosol generating substrate has a substantially flat surface and is transported away from the continuous web **34** of aerosol generating substrate **10**, for example in an upward direction as best seen in FIG. **7a**, by transport rollers **92**, **94** so that the continuous strip **218** and the continuous web **34** can be processed separately by the apparatus **230**.

The apparatus **230** also comprises a susceptor supply reel **38** (e.g. a second bobbin) which carries a continuous web **40** of susceptor material, feed rollers **42**, **44** for controlling the

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feed of the continuous web **40** of susceptor material, an adhesive applicator unit **46**, a first susceptor cutting unit **48** and a second susceptor cutting unit **49**.

The apparatus **230** further comprises an optional heater **50**, feed rollers **51**, a strip folding unit **55**, a strip cutting unit **52**, feed rollers **54**, a rod forming unit **56**, and a rod cutting unit **58**.

Susceptor Strip Preparation

In operation, a continuous web **34** of aerosol generating substrate **10** is continuously supplied from the substrate supply reel **32** and a continuous strip **218** of aerosol generating substrate **10** is separated from an edge **19** of the continuous web **34** by the rotary cutter unit **290** and transported away from the continuous web **34** by the transport rollers **92**, **94** as described above. At the same time, a continuous web **40** of susceptor material is continuously supplied from the susceptor supply reel **38**, via the feed rollers **42**, **44**, to the adhesive applicator unit **46**. The adhesive applicator unit **46** applies an adhesive **47** to a surface of the continuous web **40** of susceptor material. In the illustrated example, the adhesive applicator unit **46** applies the adhesive **47** to the surface of the continuous web **40** of susceptor material intermittently, and across the full width of the web **40**. In this way, discrete adhesive areas **60** (see FIGS. **3** and **8**) are formed on the surface of the continuous web **40** of susceptor material, with adhesive-free areas **62** being formed between adjacent adhesive areas **60** in the direction of travel of the continuous web **40** of susceptor material.

The continuous web **40** of susceptor material is supplied from the adhesive applicator unit **46** to the first susceptor cutting unit **48** which continuously cuts the continuous web of susceptor material to form a plurality of susceptor patches **28** (see FIG. **8**). The construction and operation of the first susceptor cutting unit **48** is the same as that described above in connection with FIG. **5**.

The second susceptor cutting unit **49** is positioned downstream of the first susceptor cutting unit **48** and cuts each susceptor patch **28** into a plurality of susceptor strips **18**, in the illustrated embodiment three susceptor strips **18** as best seen in FIG. **7b**. As will become apparent from the description below, each susceptor strip **18** corresponds to an elongate second strip **13** (i.e., the elongate susceptor **12**) in the finished aerosol generating article **1** described above with reference to FIGS. **1a** and **1b**.

The continuous web **40** of susceptor material, and hence the susceptor patches **28**, have a width which is less than a width of the continuous strip **218** of aerosol generating substrate **10**. For example, the continuous web **40** of susceptor material, and hence the susceptor patches **28**, can have a width of between approximately 0.1 mm and 7 mm. In some embodiments, the susceptor patches **28** can have a length of between approximately 5 mm and 50 mm in the direction of travel of the continuous web **40** of susceptor material and can have a thickness of between approximately 1 µm and 500 µm.

In order to minimise soiling of the first susceptor cutting unit **48** by the adhesive **47** applied to the continuous web **40** of susceptor material by the adhesive applicator unit **46**, the first susceptor cutting unit **48** cuts the continuous web **40** of susceptor material in the adhesive-free areas **62**, that is at positions between the adhesive areas **60** on the surface of the continuous web **40** of susceptor material (see FIG. **8**). This can be achieved by synchronising the operation of the first susceptor cutting unit **48** with the movement of the continuous web **40** of susceptor material.

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## Susceptor Strip Application

The susceptor strips **18** provided by the second susceptor cutting unit **49** can be applied to the flat surface of the continuous strip **218** of aerosol generating substrate **10** so that there is a constant and predetermined spacing **74** between the edges of each successive set of susceptor strips **18** in the longitudinal direction (i.e., the direction of travel) of the continuous strip **218**, for example as shown in FIGS. **7b** and **8**, and so that there is a small lateral spacing between the susceptor strips **18** in the transverse (width) direction of the continuous strip **218** as shown in FIG. **7b**. The constant and predetermined spacing **74** between the edges of the susceptor strips **18** may, for example, be between 1.0 mm and 20 mm. The constant and predetermined spacing **74** may be approximately mm. The constant and predetermined spacing **74** is achieved in the same manner described above in connection with the apparatus **30** and corresponding method.

The susceptor strips **18** with the adhesive **47** applied thereto are continuously and consecutively adhered to the flat surface of the continuous strip **218** of aerosol generating substrate **10** substantially along a centre of the continuous strip **218**. Adjacent sets of susceptor strips **18** are spaced apart in the direction of travel of the continuous strip **218** of aerosol generating substrate **10** by the constant and predetermined spacing **74** between the edges of the susceptor patches **28** that is generated when the susceptor patches **28** are formed in the first susceptor cutting unit **48**.

In order to ensure that there is adequate adhesion between the susceptor strips **18** and the substantially flat surface of the continuous strip **218** of aerosol generating substrate the susceptor strips **18** can be pressed onto the substantially flat surface by a cam roller **76**, shown diagrammatically in FIG. **7a**. The rotation of the cam roller **76** is synchronized with the movement of the continuous strip **218** of aerosol generating substrate **10** so that a pressing force is applied to consecutive susceptor strips **18**, but not to the spaced regions between consecutive susceptor strips **18**.

Depending on the properties of the adhesive **47** applied to the continuous web **40** of susceptor material (and hence to the susceptor strips **18**) by the adhesive applicator unit **46**, the continuous strip **218** of aerosol generating substrate **10** and the susceptor strips **18** adhered to the surface thereof can be heated by the optional heater **50**. As noted above, this may help to cure or set the adhesive **47**, and thereby ensure a good bond between the susceptor strips **18** and the flat surface of the continuous strip **218** of aerosol generating substrate **10**.

## Strip Cutting

After the continuous strip **218** of aerosol generating substrate **10** has been separated from an edge **19** of the continuous web **34** of aerosol generating substrate **10** by the rotary cutter unit **290**, the remaining web **34** of aerosol generating substrate **10** is fed to the strip cutting unit **52** (best seen in FIG. **9**). The strip cutting unit **52** cuts the continuous web **34** of aerosol generating substrate **10** across its full width to form a plurality of continuous aerosol generating strips **16** which correspond to the elongate first strips **15** in the finished aerosol generating article **1** described above with reference to FIGS. **1a** and **1b**. In an embodiment, the strip cutting unit **52** cuts the continuous web **34** of aerosol generating substrate **10** to form aerosol generating strips **16** having a strip width of approximately 1 mm.

As shown in FIGS. **7a** and **9**, the strip cutting unit **52** is a rotary cutter unit **78** and comprises first and second cutting drums **80**, **82**. The first cutting drum **80** includes circumferentially extending first cutting formations **84** and the second

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cutting drum **82** includes circumferentially extending second cutting formations **86**. The first and second cutting formations **84**, **86** cooperate (e.g. intermesh) to shear cut the continuous web **34** of aerosol generating substrate **10** in the direction of travel of the continuous web **34** to form the plurality of aerosol generating strips **16**, and specifically to form the elongate first strips **15** illustrated in FIGS. **1a** and **1b**.

## Strip Folding and Rod Formation

The elongate first strips (i.e., the aerosol generating strips **16**) formed by cutting the continuous web **34** of aerosol generating substrate **10** are conveyed to the rod forming unit **56** where they are formed into a continuous rod **88**.

The continuous strip **218** of aerosol generating substrate **10** (i.e., the elongate carrier strip **17**) with the adhered susceptor strips **18** (i.e., the elongate second strips **13**) is conveyed by the feed rollers **51** to the strip folding unit **55** which folds the elongate carrier strip **17** along fold lines **7** to form the Z-shaped carrier strip **17** described above having elongate carrier regions **17a**, **17b**, **17c** to which the elongate second strips **13** (i.e., the susceptor strips **18**) are adhered. After that, the Z-shaped carrier strip **17** with the adhered elongate second strips **13** is conveyed to the rod forming unit **56** and is combined with the elongate first strips **15** (i.e., the aerosol generating strips **16**) to form the continuous rod **88**.

If desired, a continuous sheet of wrapping paper (not shown) can be supplied to the rod forming unit **56** from a supply reel (not shown) or can be supplied to a separate wrapping unit (again from a supply reel) which can be positioned downstream of the rod forming unit **56**. As the sheet of wrapping paper is transported and guided through the rod forming unit **56** or the separate wrapping unit, it can be wrapped around the elongate first strips **15**, the Z-shaped carrier strip **17** and the adhered elongate second strips **13** so that the continuous rod **88** is circumscribed by a wrapper **14**. Rod Cutting

The continuous rod **88** (optionally circumscribed by a wrapper **14**) is then transported to the rod cutting unit **58** where it is cut at appropriate positions into predetermined lengths to form multiple aerosol generating articles **1**. The aerosol generating articles **1** formed by the rod cutting unit **58** may have a length between 5.0 mm and 50 mm, possibly between 10 mm and 30 mm. The aerosol generating articles **1** formed by the rod cutting unit **58** may have a length of 20 mm. It will be understood that this length corresponds to the length of the aerosol generating substrate **10** described above with reference to FIGS. **1a** and **1b**. The continuous rod **88** is preferably cut repeatedly by the rod cutting unit **58** substantially at a midpoint between the edges of the longitudinally spaced elongate second strips **13** (i.e., the susceptor strips **18**). In this way, the elongate second strips **13** are not cut by the rod cutting unit **58**, thereby reducing wear on the cutting elements. Further, because the elongate second strips **13** are shorter than the aerosol generating strips **16**, the ends of the elongate second strips **13** (i.e., the susceptor strips **18**) are not visible at either end of the aerosol generating articles **1** formed by the rod cutting unit **58**. It will be understood that this type of method is particularly suitable for the mass production of aerosol generating articles **1**.

## Final Assembly

Further units (not shown) may be arranged downstream of the rod cutting unit **58** and may be configured to provide one or more additional components such as the mouthpiece segment **20** described above and to assemble these with the individual aerosol generating articles **1** formed by the rod cutting unit **56** to form finished aerosol generating articles **1**, for example of the type illustrated in FIG. **1**. In this case, a

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separate wrapping unit may be provided downstream of the rod cutting unit 58 so that the assembled components can be simultaneously wrapped to form the finished aerosol generating articles 1. The further units may form part of the apparatus 230 or may be separate, stand-alone, units forming part of a final assembly line.

Although exemplary embodiments have been described in the preceding paragraphs, it should be understood that various modifications may be made to those embodiments without departing from the scope of the appended claims. Thus, the breadth and scope of the claims should not be limited to the above-described exemplary embodiments.

Any combination of the above-described features in all possible variations thereof is encompassed by the present disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to”.

The invention claimed is:

1. An aerosol generating article comprising:
  - a plurality of elongate first strips comprising an aerosol generating material;
  - a plurality of elongate second strips comprising an inductively heatable susceptor material; and
  - at least one elongate carrier strip to which each of the plurality of elongate second strips is attached;
 wherein:
  - the at least one elongate carrier strip is folded along one or more fold lines to form two or more elongate carrier regions, and one or more of the plurality of elongate second strips is attached to each elongate carrier region, and
  - the elongate first strips, the elongate second strips and the at least one elongate carrier strip are arranged to form a rod-shaped aerosol generating article.
2. The aerosol generating article according to claim 1, wherein one or more of the elongate second strips is adhered to each elongate carrier region.
3. The aerosol generating article according to claim 1, wherein the at least one elongate carrier strip is a Z-shaped carrier strip.
4. The aerosol generating article according to claim 1, wherein each of the plurality of elongate first strips has a width which is less than a width of each of the plurality of elongate second strips.
5. The aerosol generating article according to claim 1, wherein each of the plurality of elongate first and second strips has a distal end, the distal ends of the elongate first

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strips form a distal end of the aerosol generating article, and the distal ends of the elongate second strips are positioned inwardly from the distal ends of the elongate first strips so that the distal ends of the elongate second strips are not visible at the distal end of the aerosol generating article.

6. The aerosol generating article according to claim 1, wherein a length of each of the elongate second strips is less than a length of each of the elongate first strips.

7. The aerosol generating article according to claim 1, wherein a length of the at least one elongate carrier strip is equal to a length of each of the elongate first strips.

8. The aerosol generating article according to claim 1, wherein the at least one elongate carrier strip comprises an aerosol generating material.

9. The aerosol generating article according to claim 1, further comprising a filter segment at a proximal end of the aerosol generating article and at least one tubular segment upstream of the filter segment.

10. The aerosol generating article according to claim 1, wherein the elongate first strips have a plurality of different orientations within a cross-section of the rod-shaped aerosol generating article.

11. The aerosol generating article according to claim 1, wherein each of the plurality of elongate second strips has a thickness between 1  $\mu\text{m}$  and 500  $\mu\text{m}$ .

12. An aerosol generating article according to claim 1, wherein each of the plurality of elongate first strips has a length between 10 mm and 30 mm.

13. An aerosol generating article according to claim 1, wherein each of the plurality of elongate first strips has a thickness between 150  $\mu\text{m}$  and 300  $\mu\text{m}$ .

14. An aerosol generating article according to claim 1, wherein the aerosol generating material comprises a tobacco material.

15. An aerosol generating article according to claim 1, wherein the inductively heatable susceptor material comprises a metal.

16. The aerosol generating article according to claim 11, wherein each of the plurality of elongate second strips has a thickness between 10  $\mu\text{m}$  and 100  $\mu\text{m}$ .

17. An aerosol generating article according to claim 12, wherein each of the plurality of elongate first strips has a length of approximately 20 mm.

18. An aerosol generating article according to claim 13, wherein each of the plurality of elongate first strips has a thickness of approximately 220  $\mu\text{m}$ .

19. An aerosol generating article according to claim 15, wherein the metal comprises a stainless steel or a carbon steel.

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